Practice Quiz: Systems (Neuroscientific Frontiers)

Name: __ ***Date:*** __

Part A: Multiple Choice

1. In the cortical hierarchy, feedforward connections primarily carry: A) Top-down predictions B) Bottom-up prediction errors — mismatches between predictions and sensory data C) Global modulatory signals D) Motor commands

2. Feedback connections originate from which cortical layers? A) Layers 2/3 B) Layers 5/6 — deep pyramidal cells generating top-down predictions C) Layer 4 only D) Layer 1 only

3. In the canonical microcircuit, superficial pyramidal cells (L2/3) compute: A) Motor commands B) Prediction errors — sent feedforward to the next hierarchical level C) Top-down expectations D) Neuromodulatory signals

4. The basal ganglia implement which Active Inference computation? A) Sensory prediction errors B) Policy selection — evaluating actions by expected value and selecting the best policy C) Forward motor models D) Episodic memory

5. The cerebellum functions as a: A) Memory storage system B) Forward model for motor control — predicting sensory consequences of actions C) Language center D) Emotion processor

6. Acetylcholine in Active Inference primarily controls: A) Reward B) Precision of sensory prediction errors — relating to attention and learning rate for new sensory data C) Sleep D) Motor speed

7. The thalamus functions as a: A) Simple relay B) Precision gate — controlling which prediction errors reach cortex by modulating their gain C) Memory store D) Motor center

Part B: Essay Questions

1. Trace the neural processing of a specific perceptual experience (e.g., hearing your name in a noisy room) through the cortical hierarchy. For each level, specify the predictions generated, the prediction errors computed, and how precision weighting (attention) modulates the process. Include at least one subcortical contribution. (400 words)

2. Compare the canonical microcircuit model (Bastos et al., 2012) with alternative neural coding frameworks (e.g., divisive normalization, sparse coding). What unique predictions does the predictive coding canonical microcircuit make? What empirical tests could distinguish these accounts? (400 words)

3. Explain how psychiatric conditions can be understood as disorders of precision weighting, using the neuromodulatory framework. Choose two conditions (e.g., schizophrenia, depression, autism, anxiety) and explain each as aberrant precision in a specific neuromodulatory system. What therapeutic interventions does this perspective suggest? (400 words)